

# Short Intro to Decomposition Methods

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# Preliminaries

- Brief introduction of the instructor
- Slides and lab(s) available on Github
- `https://github.com/benjisamschlu/decomposition_methods_berkeley26.git`
- Student introductions/familiarity with demography

# Outline

- What is Demography?
- Aggregate measures
- Standardization and its limits
- Kitagawa decomposition
- Life tables
- Arriaga decompostion

# What is Demography?

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*"Demography is the scientific study of human population, including its size, distribution, composition, and the factors that determine changes in its size, distribution, and composition."*(Shryock and Siegel 2004)

*"Quantitative study of groups of people."*(Lutz 2021)

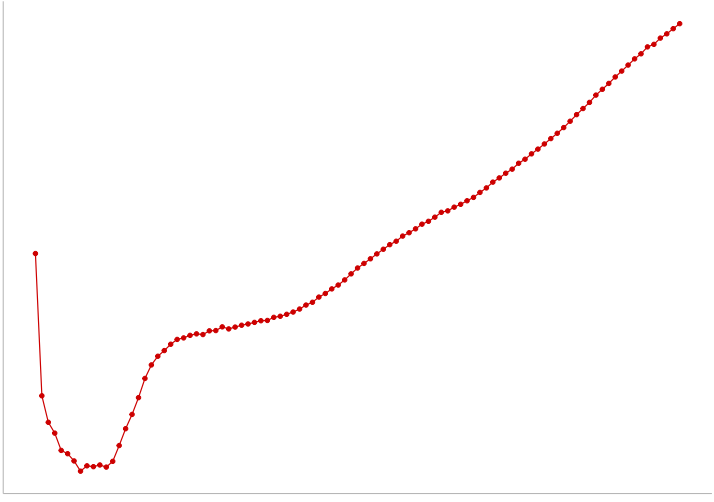
# Demographic Balancing Equation

$$P(t+n) = P(t) + \ldots\ldots\ldots - \ldots\ldots\ldots + \ldots\ldots\ldots - \ldots\ldots\ldots \quad (1)$$

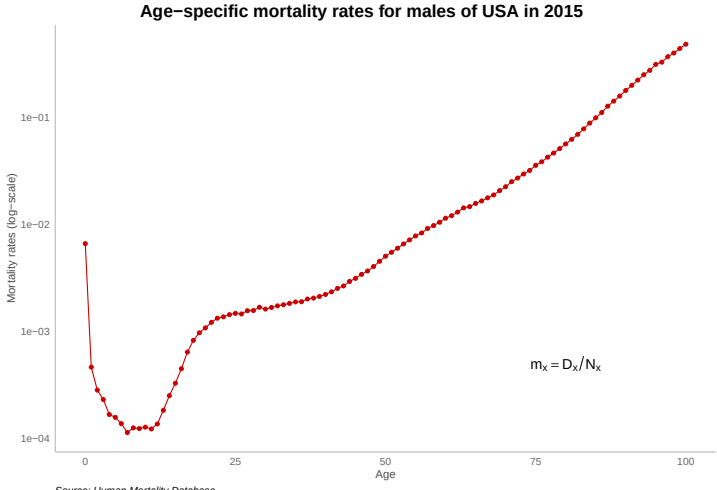
# Demographic Balancing Equation

$$P(t+n) = P(t) + B[t, t+n] - D[t, t+n] + I[t, t+n] - O[t, t+n] \quad (2)$$

# What are we looking at?

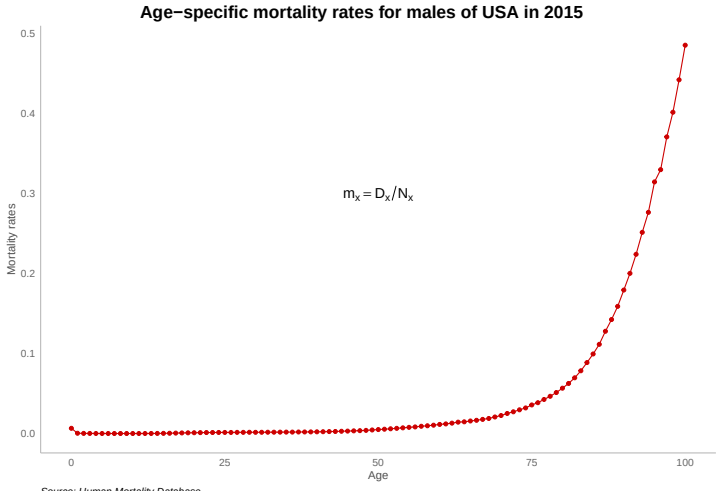


# Mortality age schedule



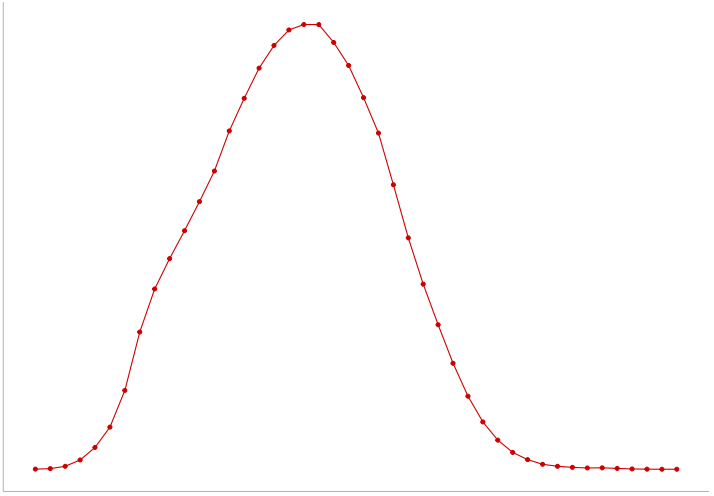

*Source: Human Mortality Database*

# Mortality age schedule

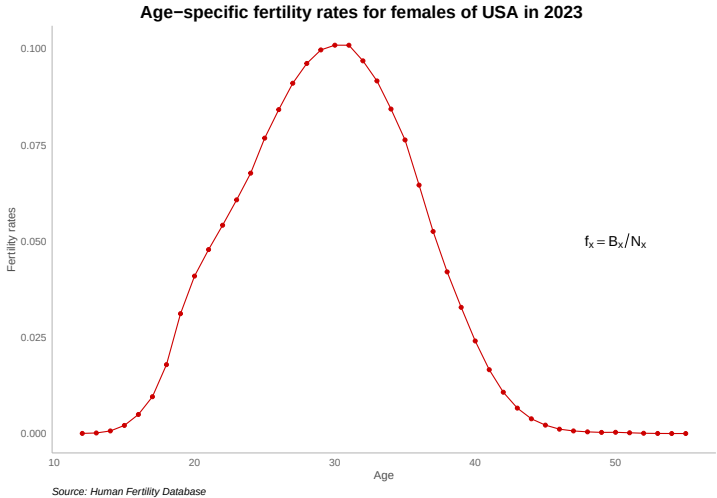



*Source: Human Mortality Database*

# What are we looking at?



# Fertility age schedule



# Aggregate measures

- Summary measures of population change due to births, deaths or migration
  - Crude birth rates (?), general fertility rate (?), crude death rate (?), crude net migration rate (?), ...

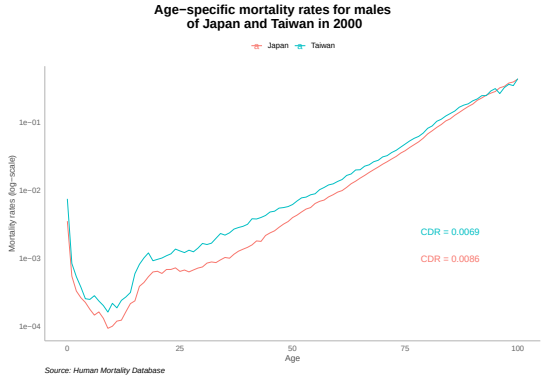
# Aggregate measures

- Summary measures of population change due to births, deaths or migration
  - Crude birth rates (Bth/N), general fertility rate (Bth/$N^f_{15-49}$), crude death rate (Dth/N), crude net migration rate (I-O/N), ...

# Aggregate measures

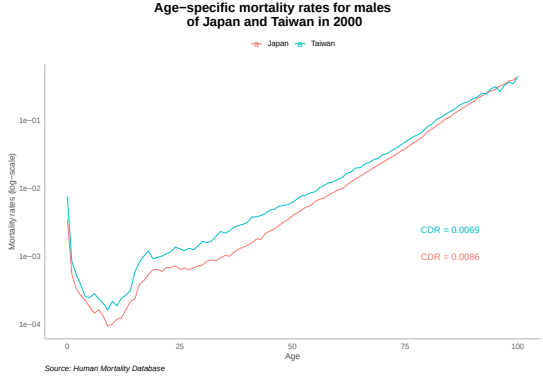
- Summary measures of population change due to births, deaths or migration
  - Crude birth rates (Bth/N), general fertility rate (Bth/$N^f_{15-49}$), crude death rate (Dth/N), crude net migration rate (I-O/N), ...
- We compare these measures across populations and over time

# Crude death rate




Anything surprising?

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Anything surprising?

- **Rate effects** and **compositional effects**

## Re-expressing crude death rate

$$CDR = \frac{D}{N} = \frac{\sum_x ?}{N} \tag{3}$$

(5)

(7)

(8)

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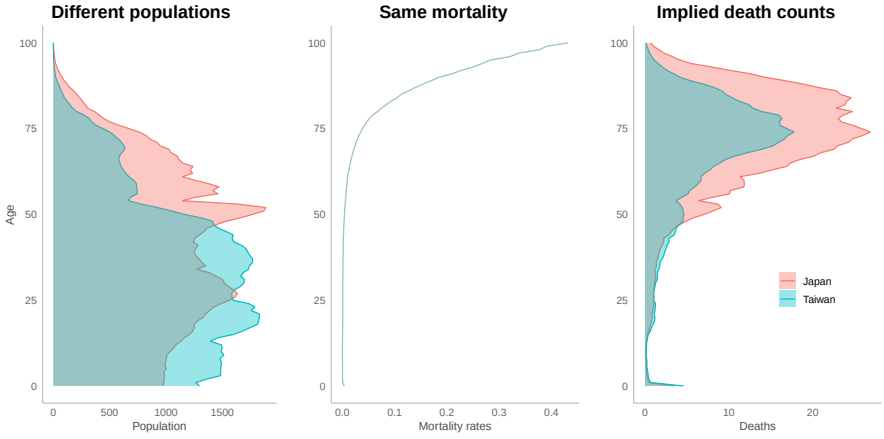
$$= \sum_x M_x \frac{N_x}{N} \tag{5}$$

$$= \sum_x \underbrace{M_x}_{\text{Rates}} \underbrace{C_x}_{\text{Comp.}} \tag{6}$$

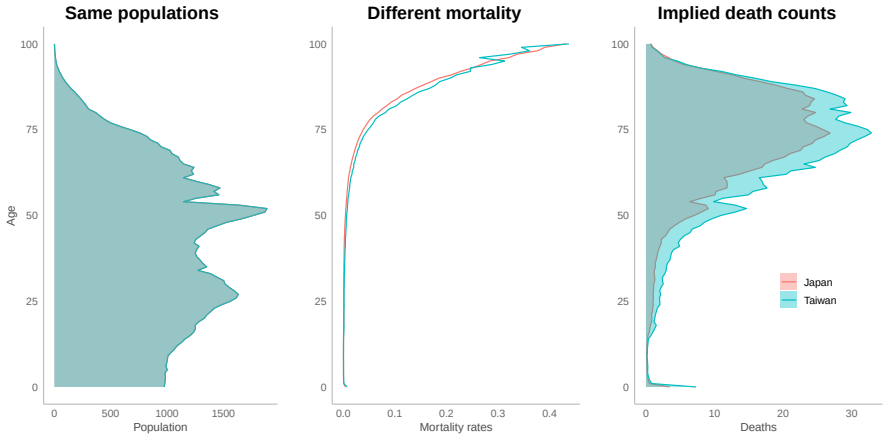
$$= \sum_x w_x M_x \tag{7}$$

$$\tag{8}$$

# Compositional effects



# Rate effects



## Difference between crude rates

$$\Delta CDR = CDR^{\text{Twn}} - CDR^{\text{Jpn}} \tag{9}$$

$$= \sum_x M_x^{\text{Twn}} C_x^{\text{Twn}} - \sum_x M_x^{\text{Jpn}} C_x^{\text{Jpn}} \tag{10}$$

$$= f(\Delta\text{rates}, \Delta\text{composition}) \tag{11}$$

## Standardization methods

**Direct standardization:** Remove compositional effect and focus on the difference in rates.

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How? Compare $CDR^{Jpn}$ with

$$CDR_{\text{Std}}^{\text{Twn}} = \sum_x M_x^{\text{Twn}} C_x^{\text{Jpn}} \tag{12}$$

## Standardization methods

**Direct standardization:** Remove compositional effect and focus on the difference in rates.

How? Compare $CDR^{Jpn}$ with

$$CDR_{\text{Std}}^{\text{Twn}} = \sum_x M_x^{\text{Twn}} C_x^{\text{Jpn}} \tag{12}$$

**Indirect standardization:**

$$SMR^{\text{Twn}} = \frac{\sum_x M_x^{\text{Twn}} C_x^{\text{Twn}}}{\sum_x M_x^{\text{Jpn}} C_x^{\text{Twn}}} = \frac{D^{\text{Twn}}}{\sum_x M_x^{\text{Jpn}} C_x^{\text{Twn}}} = \frac{D^{\text{Twn}}}{E^{\text{Twn}}} \tag{13}$$

# Standardization: a partial solution

Limitations?

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Limitations?

- Choice of standard
- Dependence on standard
- Components contribution

# Decomposition: improve comparison methods

What drives the difference in aggregate measures?

- Rates vs composition effect
- Contribution of each effect to difference in crude rates

# Kitagawa decomposition

Kitagawa 1955 (see Appendix):

$$\Delta CDR = \underbrace{\sum_x \left( \frac{M_x^{Twn} + M_x^{Jpn}}{2} \right) \left( \frac{N_x^{Twn}}{N^{Twn}} - \frac{N_x^{Jpn}}{N^{Jpn}} \right)}_{\text{Difference due to pop. composition}} + \underbrace{\sum_x \left( \frac{\frac{N_x^{Twn}}{N^{Twn}} + \frac{N_x^{Jpn}}{N^{Jpn}}}{2} \right) \left( M_x^{Twn} - M_x^{Jpn} \right)}_{\text{Difference due to rates}} \quad (14)$$

# Kitagawa decomposition

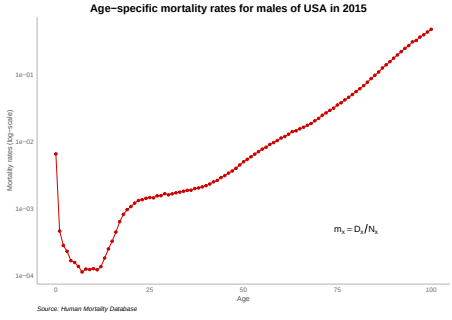
- **Composition effect**: difference in age composition, holding rates at the average of the two populations
  - How much of the difference in CDR is due to the population being younger of older?
- **Rate effect**: difference in mortality rates, holding the age composition at the average of the two populations
  - How much of the difference in CDR is due to the mortality rates being different?
- The effects can be greater or smaller than 0. The total difference in CDR is the sum of the two effects.

# Life tables - machinery

Main input?

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Main input?




Convert ${}_nm_x$ into ${}_nq_x$ (${}_na_x$!)

# Life tables - summary measures

Who is the star of the life table?

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- Life expectancy

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Expectation matters but what about the spread?

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Who is the star of the life table?

- Life expectancy

Expectation matters but what about the spread?

- Lifespan disparity

# Life tables - cohort vs period

- What is the difference (think about the main input)?

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- Why do we use period life tables?

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- What is the difference (think about the main input)?
- Why do we use period life tables?
  - Synthetic cohort
- Impact of Covid19: excess mortality

# Arriaga decomposition

Arriaga 1984:

- Comparing life expectancies across populations and over time
- Contribution of each age group in life expectancy difference
  - Age group dependence: direct vs indirect effect

## Lab

We will use data from the Human Mortality Database (HMD) and the Human Fertility Database (HFD).

**Section A**

2000 vs 2020, Japan females. Compute

1. CDR
2. SMR
3. Kitagawa decomposition for CDR

**Section B**

Chile vs Ireland females, 2020. Compute

1. GFR
2. Adapt Kitagawa decomposition for GFR

# References I


- Arriaga, Eduardo E (1984). "Measuring and explaining the change in life expectancies". In: *Demography* 21.1, pp. 83–96.
- Kitagawa, Evelyn M (1955). "Components of a difference between two rates". In: *Journal of the american statistical association* 50.272, pp. 1168–1194.
- Lutz, Wolfgang (2021). *Advanced introduction to demography*. Edward Elgar Publishing.
- Shryock, Henry S. and Jacob S. Siegel (2004). *The Methods and Materials of Demography*. Ed. by Jacob S. Siegel and David A. Swanson. 2nd ed. San Diego, CA and London: Elsevier Academic Press.

# Appendix - Kitagawa (1955)

$$\Delta CDR = \underbrace{\sum_x M_x^{Jpn}\left[\frac{N_x^{Twn}}{N^{Twn}} - \frac{N_x^{Jpn}}{N^{Jpn}}\right]}_{\text{Change in } x\text{-composition}} +$$

$$\underbrace{\sum_x \frac{N_x^{Jpn}}{N^{Jpn}}\left[M_x^{Twn} - M_x^{Jpn}\right]}_{\text{Change in rates using pop 1 as the standard}} + \tag{15}$$

$$\underbrace{\sum_x \left(M_x^{Twn} - M_x^{Jpn}\right)\left[\frac{N_x^{Twn}}{N^{Twn}} - \frac{N_x^{Jpn}}{N^{Jpn}}\right]}_{\text{Interaction between rates and composition}}$$